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import java.util.*;
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public class Main {
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    public static String Encryption() {

        Scanner sc = new Scanner(System.in);
        int key;
        char[] text;

        while (true) {
            System.out.print("Enter the Text in LowerCase : ");

            String str = sc.nextLine();
            text = str.toCharArray();

            boolean valid = true;

            for (int i = 0; i < text.length; i++) {

                if (text[i] == ' ') {
                    continue;
                }
                else if (text[i] < 'a' || text[i] > 'z') {
                    valid = false;
                    break;
                }
            }

            if (!valid) {
                System.out.println(
                    "Please Enter text only and also make sure it is in lower Case:"
                );
                continue;
            }
            else {
                break;
            }
        }

        System.out.print("Enter the Key : ");
        key = sc.nextInt();

        char cypher[] = new char[text.length];

        for (int i = 0; i < text.length; i++) {

            if (text[i] == ' ') {
                cypher[i] = text[i];
                continue;
            }
        }
    }
}
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        int x = text[i] - 97;

        cypher[i] = (char) (((x + key) % 26) + 97);
    }

    String cyphertext = new String(cypher);
    cyphertext = cyphertext.toUpperCase();

    return cyphertext;
}

public static String Decryption() {

    Scanner sc = new Scanner(System.in);
    int key;
    char[] text;

    System.out.print("Enter The Encrypted text In UpperCase: ");

    String str = sc.nextLine();
    str = str.toLowerCase();

    text = str.toCharArray();

    System.out.print("Enter The Key: ");
    key = sc.nextInt();

    char cypher[] = new char[text.length];

    for (int i = 0; i < text.length; i++) {

        if (text[i] == ' ') {
            cypher[i] = text[i];
            continue;
        }

        int m = text[i] - 97;

        cypher[i] = (char) (((m - key) + 26) % 26) + 97);
    }

    String cyphertext = new String(cypher);
    cyphertext = cyphertext.toLowerCase();

    return cyphertext;
}

public static int Brute_force(char x, char y) {

    int i = 1;

    x = Character.toLowerCase(x);

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while (true) {

    if (((x - 97) - i) % 26) + 26 == y) {
        return i;
    }
    else {
        i++;
    }

    if (i > 26) {
        return 0;
    }
}

}

public static void main(String s[]) {

    while (true) {

        Scanner sc = new Scanner(System.in);

        System.out.print(
            "For Encryption : 1 " +
            "\t For Decryption : 2" +
            "\nFor Bruteforce : 3" +
            "\t For Exit : 99" +
            "\nEnter Your Choice : "
        );

        int a = sc.nextInt();

        if (a == 1) {

            System.out.println(
                "This is The Encrypted Text : " + Encryption()
            );
        }
        else if (a == 2) {

            System.out.println(
                "This is Decrypted Text : " + Decryption()
            );
        }
        else if (a == 3) {

            System.out.print(
                "We Have two Option 1st is Key Finding : 1 " +
                " \tSecond Is UnAvailable \n"
            );

            System.out.println(
                "So Directly We are using 1st option : "
            );
        }
    }
}

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        System.out.print(
            "So First Give the Encrypted Key: "
        );

        char x = sc.next().charAt(0);

        System.out.print(
            "Enter The Decrypted Code :"
        );

        char y = sc.next().charAt(0);

        int r = Brute_force(x, y);

        if (r == 0) {
            System.out.print(
                "The Key is bigger then 26"
            );
        }

        System.out.println(
            "The key is :" + r
        );
    }
    else if (a == 99) {
        break;
    }
}
}
}

```

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}

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